Reg. No. | | | | | | | | | | | [14ME34]

THIRD SEMESTER B.TECH EXAMINATION, DECEMBER–2015

APPLIED MECHANICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions choosing at least **TWO** questions from each part.
2. Assume any data necessary and indicate.

PART-A

1. a. Define the following: **04**

- **i.** True stress
- **ii.** Poission's ratio
- **iii.** True strain
- **iv.** Factor of safety

b. A Specimen of steel 25mm diameter with a gauge length of 200mm is tested to destruction. It has an extension of 0.16mm under a load of 80KN and the load at elastic limit is 160KN. The maximum load is 180KN. The total extension is 56mm and diameter at neck is 18mm. Find **10**

- **i.** The stress at elastic limit.
- **ii.** Young's Modulus.
- **iii.** Percentage elongation.
- **iv.** Percentage reduction in area.
- **v.** Ultimate tensile stress.

2. a. Explain principle of super position. **04**

b. A compound bar of length 500mm consists of a strip of aluminium 50mm wide X 20mm thick and a strip of steel 50mm wide X 15mm thick rigidly joined at ends. If the bar is subjected to a load of 50KN, find the stresses developed in each material and the extension of the bar. Take elastic modules of aluminium and steel as $1X10^5 N/mm^2$ and $2X10^5 N/mm^2$ respectively. The compound bar is shown in Fig 2(a). **04**

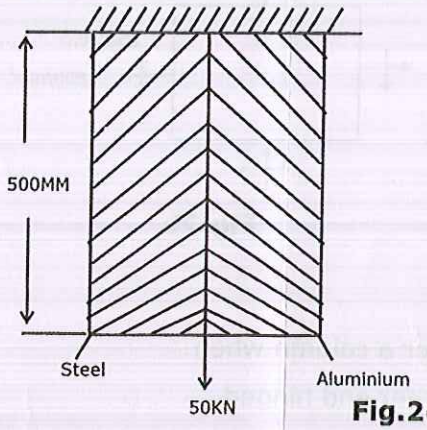

Fig.2(a)

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3. The simply supported beam as shown in fig-3 carries two concentrated loads and **14**

a uniformly distributed load. Draw the shear force diagram and bending moment diagram.

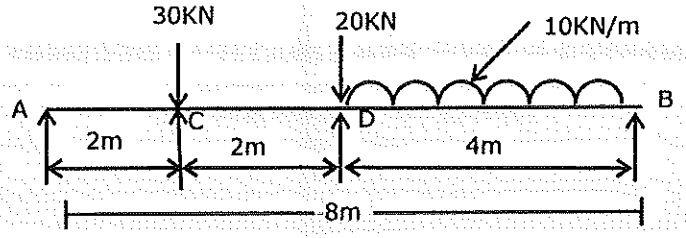

Fig-3

4. State the assumptions made in bending theory and derive the bending equation. **14**

PART-B

5. Derive moment-curvature relationship for deflection of beam. **14**

6. **a.** Define the following: **04**

- **i.** Torsional rigidity
- **ii.** Polar modulus
- **iii.** Polar moment of Inertia
- **iv.** Twisting moment

b. A hallow circular shaft of 6m length and inner and outer diameter of 75mm and 100mm is subjected to torque of 10KNm. If G=80GPa, determine shear stress produced and the total angle of twist. **10**

7. **a.** Explain the following: **04**

- **i.** Principal stress
- **ii.** Principal plane

b. The state of stress at a point in a strained material is as shown in fig7(b). Determine **10**

- **i.** The direction of the principal plane.
- **ii.** The magnitude of principal stresses.
- **iii.** The magnitude of the maximum shear stress.

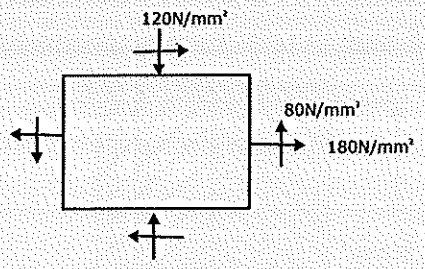

Fig-7b

8. Derive the Euler's formula for a column when **7+7**

- **i.** One end fixed and other end hinged.
- **ii.** Both ends fixed.